Project progress 10-04-2026

Rolfe • April 10, 2026 • Estimated reading time: 2 min

What

This week I have been implementing a rate limit feature to the application. To prevent users from spamming requests in a very short period and also trying to login in a short time frame with wrong login credentials. I also started moving my files across different folders to have some structure in place. This caused me to have to go through each file and manually change the file path in the hyperlinks. This here showed that the way I wrote the application until now is not scalable. As I was checking the code I realised that I was not following design patterns, I had pieces of code that do perform the same action in multiple files.

So What

So to avoid having to do repetitive task such as renaming the file path in each hyperlink, I should have created a template folder for the navigation bar and the head of the document, these 2 were sections I had to rewrite the most across the files. Having this template folder would allow me or someone else to make changes on the templates and apply them to all the files, saving time. For the repetitive pieces of code, I will now replace them with functions, such as the if statements I used to authenticate the user or the sql queries that select the user account. This process took time from other parts of the project that required more attention, I should have done this from the beginning

Now What

This experience has showed me the importance of scalability in a program. Now I will follow design patterns more closely, and establish a structure for the project as early as possible. I will organise the code better by following the Model-View-Controller design pattern. I will create reusable components such as shared templates and functions for repeated logic like authentication and database queries. This will allow me to reduce duplication, improve maintainability, and avoid time-consuming refactoring later on.